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```
clc; clear; close all;

%% FIR FILTER PARAMETERS
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];
n = 0:N-1;

%% 1. DIRECT REALIZATION
yd = filter(b,1,x);

%% 2. CASCADE REALIZATION
z = roots(b);
g = b(1);
yc = g*x;

for k = 1:length(z)
    yc = filter([1 -z(k)],1,yc);
end

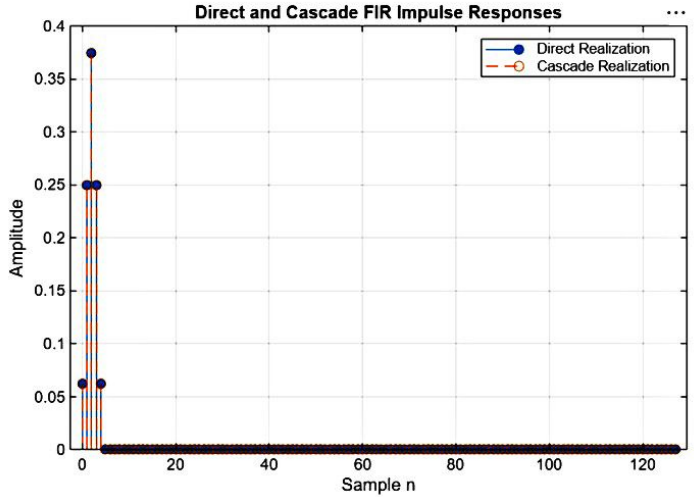
yc = real(yc);

%% 3. SYMMETRY AND LINEAR PHASE
symErr = max(abs(b-fliplr(b)));
D = (length(b)-1)/2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end

%% 4. FREQUENCY, PHASE AND GROUP DELAY
[H,w] = freqz(b,1,1024);
```

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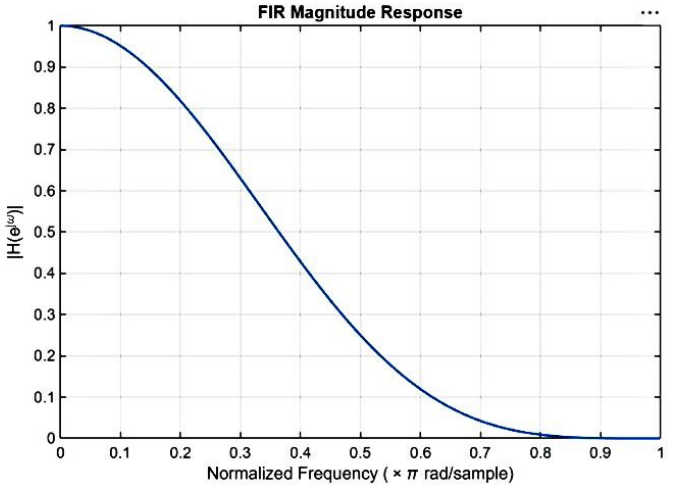
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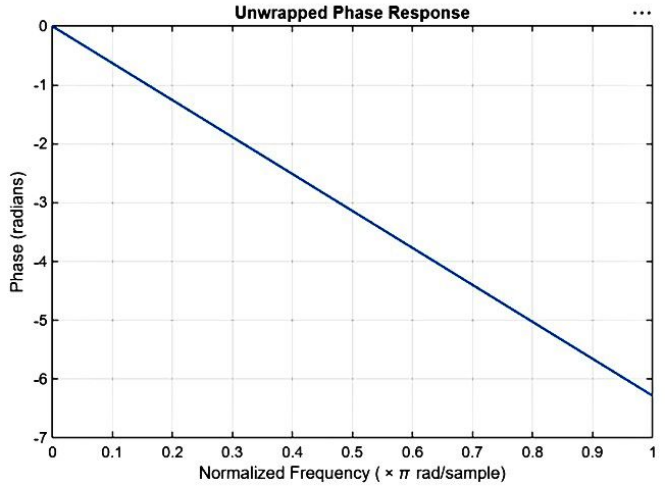
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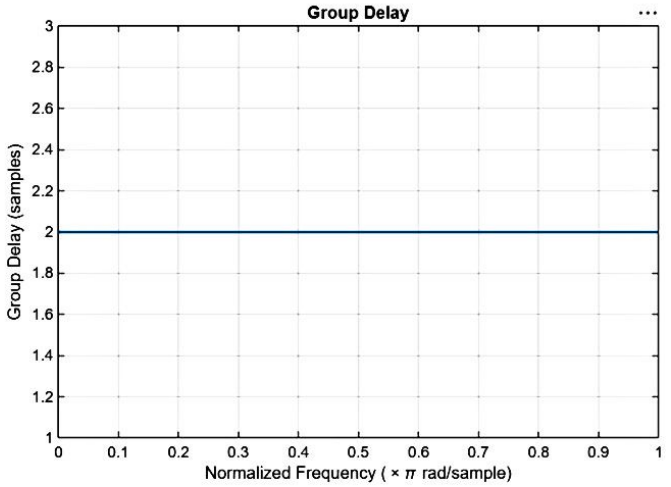
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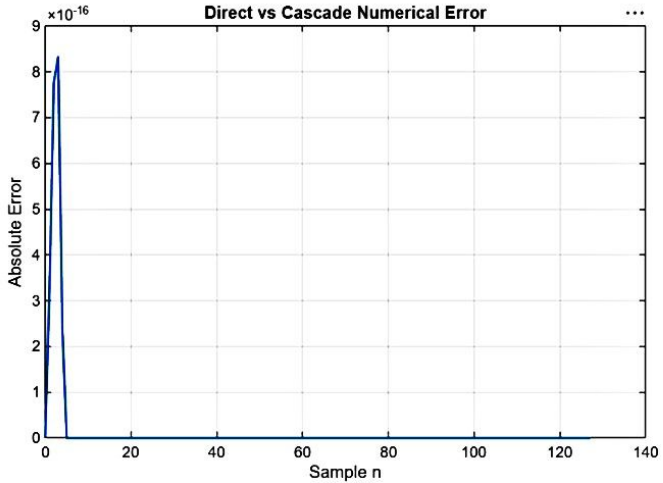
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